

# Gene Pulser® Electroprotocols

\* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

|                     |                                         |                                 |                                                                                                                                                                                  |
|---------------------|-----------------------------------------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cell Type</b>    | Mammalian, adherent                     | <b>Molecules Electroporated</b> | DNA: multiple CAT and other vectors under different promoters including HIV, SIV, human and rhesus cyto-megalovirus immediate early gene, SV40 and the simian retrovirus type 2. |
| <b>Species Used</b> | Human, HuT78, cutaneous T-cell lymphoma |                                 |                                                                                                                                                                                  |

## Before the Pulse

|                           |                                                                                       |                                |                                                                            |
|---------------------------|---------------------------------------------------------------------------------------|--------------------------------|----------------------------------------------------------------------------|
| <b>Cell Growth Medium</b> | RPMI 1640 with 10 mM dextrose and 0.1mM dithiothreitol (Sigma, GIBCO/ BRL, Flow Labs) | <b>Growth Phase at Harvest</b> | Log phase (We routinely subdivide cells 24 hours prior to electroporation) |
|                           |                                                                                       | <b>Pre-pulse Incubation</b>    | None                                                                       |
| <b>Wash Solution</b>      | Trypsin harvest, washed twice in Phosphate Buffered Saline                            |                                |                                                                            |

## The Pulse

**Instruments Used** Gene Pulser® apparatus & Capacitance

|                                    |                                                                            |                       |                           |
|------------------------------------|----------------------------------------------------------------------------|-----------------------|---------------------------|
| <b>Electroporation Temperature</b> | 25 °C                                                                      |                       |                           |
| <b>Electroporation Medium*</b>     | RPMI 1640 without Fetal Calf Serum , +10mM dextrose, 0.1 mM dithiothreitol | <b>Cuvette Gap</b>    | 0.4 cm                    |
| <b>Cell Density</b>                | 1.3 x 10 <sup>7</sup> (7) viable cells / ml                                | <b>Voltage</b>        | 0.250 kV                  |
| <b>Volume of Cells</b>             | 300 µl                                                                     | <b>Field Strength</b> | 0.625 kV/cm               |
| <b>DNA Concentration</b>           | 500 µg / ml (5 to 10 µg per pulse)                                         |                       |                           |
| <b>DNA Resuspension Buffer</b>     | Not given                                                                  | <b>Capacitor</b>      | 960 µF                    |
| <b>Volume of DNA</b>               | 10 to 20 µl                                                                | <b>Resistor</b>       | (Pulse Controller) Ω none |
| <b>After the Pulse</b>             |                                                                            | <b>Time Constant</b>  | 33 to 38 msec             |
| <b>Outgrowth Medium</b>            | DMEM culture media, 10% fetal calf serum                                   |                       |                           |

|                                       |                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Outgrowth Temperature</b>          | 37 °C                                           | <b>Relevant Publications and/or Comments</b><br><b>Note:</b> exponential values designated in parentheses. All plasmids were prepared by alkaline lysis and banded to equilibrium twice in cesium chloride gradients. Largest DNA molecule electroporated has been the rhesus cytomegalovirus genome (approx. 225 kilobases). In general, the higher the efficiency of transfection, the higher degree of cell killing. For some cell types, > 90% of the input cells are killed, although > 90% of the survivors express very high levels of input DNA. We have found that ≥ 20 µg of DNA per electroporation resulted in greatly reduced efficiency of transfection. Similarly, 2.5 µg of DNA resulted in reduced efficiency. |
| <b>Length of Incubation</b>           | 48 hours                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Selection Method or Assay Used</b> | Transient (CAT, b-gal, immunohisto- chemistry). |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Electroporation Efficiency</b>     | 50 to 100%                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Per Cent Survival</b>              | 25 to 75%                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

**Name of Submitter** Peter Barry, Ph.D., Asst. Research Virologist

**Institution Address** Univ. of California- Davis  
Department of Medical Pathology , MS1A  
Davis, CA 95616

**Survey Number**  
100